Species-Dependent Site-State Topology Specification

Legacy/Manual Stage 11E-M1: Ring-Side Anchors, Explicit Geometric Site Hypotheses, and Structural Candidate Networks

mdstats

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Contents

Species-dependent site-state topology specification

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Stage: 11E-M1 (historically labeled 11E1)

Scope

Legacy/manual Stage 11E-M1 converts certified natural-tiling geometry, persistent ring geometry, and framework semantics into a **species-specific geometric hypothesis graph**. It does not infer free-energy minima, barriers, rates, or trajectory assignments. Every physical-state hypothesis must be supplied explicitly through a `SpeciesSiteTopologyProfile`.

This specification describes the implemented explicit/manual branch. It is not the new data-driven Stage 11E1 periodic species-density estimator defined by the Stage 11 architecture manual. Public module names and release behavior are unchanged by this documentation reclassification.

This supplied-model stage is split into:

- `ring_site.py`: persistent ring-side anchors and geometric site microstates;
- `site_kinetic_network.py`: structurally admissible directed candidate paths.

Source contract

The builder requires one mutually consistent:

- `TilingGeometryCatalog`;
- `ReferenceRingGeometryCatalog`;
- `FrameworkSemanticCatalog`;
- `SpeciesSiteTopologyProfile`.

The three catalogs must carry the same tiling-geometry digest. A profile is species-specific and matches semantic interface labels and optional semantic tile labels. Unmatched ring interfaces are rejected. Absence of a bound state must be represented explicitly by `NO_BOUND_STATE`; uncertainty must be represented explicitly by `UNRESOLVED`.

Persistent ring-side anchors

Every natural-tiling window owns two topological anchors, one for each oriented ring–tile incidence. The anchors remain distinct even when a physical plane-centered state merges them. A resolved anchor retains:

- persistent window and side identity;
- adjacent tile identity and semantic label;
- ring-interface family;
- geometric O-area center;
- inward normal and right-handed in-plane axes;
- image shift relative to the canonical side-a representative.

Side-a has image shift `(0,0,0)`. Side-b has the window's `relative_tile_translation`.

Landscape taxonomy

`SiteLandscapeRegime` contains:

- `NO_BOUND_STATE`;
- `ONE_SIDED`;
- `BILATERAL_DOUBLE_WELL`;
- `PLANE_CENTERED`;
- `PLANE_OFF_CENTER_DISCRETE`;
- `PLANE_ANNULAR`;
- `GENERAL_MULTIWELL`;
- `UNRESOLVED`.

The regime is a declared geometric hypothesis, not an energetic certification.

One-sided

One state is placed a positive distance along the inward normal of the side whose adjacent tile has `active_tile_label`.

Bilateral

Two states are placed independently along the two inward normals. They remain separate physical nodes and retain their own tile exposure and image shift.

Plane-centered

One state is placed at the O-area center and references both side anchors. No false side-to-side edge is created.

Off-center discrete

`angular_count >= 2` states are placed at radius `radial_offset` in the ring plane using the canonical side-a in-plane axes and `angular_phase`. They form a cyclic set; each state retains both tile exposures.

Annular

One continuous-state placeholder is centered at the ring center and stores its annular radius. It is not discretized into artificial angular sites.

General multiwell

The profile supplies explicit `GeneralSiteTemplate` records in local `(z, rho, theta)` coordinates and a side affinity (`a`, `b`, or `plane`). No additional minima are inferred.

Optional cage-interior hypotheses

A `CageInteriorRule` may create one geometric candidate at the exact natural- tile volume centroid for every matching semantic tile. Declaring this candidate does not certify that the centroid is a metastable basin.

State placement and tile exposure

Each state stores a stable key, a Cartesian reference position, local coordinates, anchor references, and one or two `SiteTileExposure` records. An exposure records the adjacent tile index and the periodic image shift of that tile relative to the state's canonical representative.

Structural candidate network

`build_site_kinetic_network()` creates a periodic directed multigraph. Edges are structural candidates only and have no rate or barrier.

Generated edge classes are:

- `INTRA_RING_CROSSING`: the two states of a bilateral model;
- `INTRA_RING_ANGULAR`: neighboring states of a discrete angular cycle;
- `RING_TO_CAGE`: a ring-associated state and an explicitly declared cage state exposed to the same tile image;
- `INTRA_TILE_TRANSFER`: only when enabled by an explicit `TileTransferRule`.

Sharing a tile never creates an edge unless a cage rule or tile-transfer rule explicitly requests it. Directed reverse edges are stored separately. The edge translation equals target exposure shift minus source exposure shift.

Immutability and replay

All records are frozen. Mutable arrays are not exposed. Catalog and network payloads use canonical JSON and SHA-256 digests. `from_dict()` rebuilds from the supplied source catalogs and profile and rejects tampering.

Resource limits

Preflight limits bound anchors, states, edges, rules, and angular variants before bulk object construction.

Required validation

Focused tests must cover:

- two anchors per ring and opposite side geometry;
- explicit unmatched-interface rejection;
- no-bound and unresolved regimes;
- one-sided side selection by semantic tile label;
- bilateral image-labelled crossing edges;
- merging both anchors into one centered state;
- discrete angular cyclic states and edges;
- annular non-discretization;
- optional cage states and ring-to-cage paths;
- explicit-only intra-tile transfers;
- LTA profile counts derived from 24/12/16/6 semantic families;
- generic profile operation without LTA names;
- resource preflight, deterministic ordering, serialization replay, tamper rejection, and public exports.